

# Lab 4

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## Questions

1) Increasing “i\_focal\_dist” makes the image appear zoomed in, so the objects look larger and less of the scene is visible. Decreasing it makes the image appear zoomed out, so the objects look smaller and more of the scene becomes visible. This happens because a larger focal distance makes the rays more parallel to the camera’s viewing direction, giving a narrower field of view. A smaller focal distance spreads the rays farther apart and gives a wider field of view.

2) Super-sampling could be implemented by shooting several rays through slightly different positions inside each pixel instead of shooting only one ray through its centre. The positions could follow a small regular grid or use random offsets. The colours returned by all the rays would then be added together and divided by the number of samples. Pixels close to sphere edges or checkerboard boundaries would contain an average of both sides, producing smoother edges and reducing aliasing. More samples improve the result but also increase the rendering time.

3) “MAX\_DEPTH” limits the total number of rays processed. At depth 1, only the primary ray is processed, so the refractive sphere appears mostly dark. At depth 2, a transmitted ray enters the sphere but cannot exit. At depth 3, the transmitted and reflected rays are processed, but the exiting ray still does not fit in the ray array. At depth 4, the ray can leave the sphere and reach the scene behind it, making refraction visible. For a purely reflective object, depth 2 shows the first reflection and depth 3 allows another reflection bounce. After approximately depth 4 - 5, changes become very small because most rays have reached the sky or non-reflective surfaces, while the remaining rays have very low weights.